

Calm Months, Crash Months: Why Volatility Management Pays for Momentum and Not the Market

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Abstract

Built with only past data, capped leverage and trading costs, volatility management improves momentum, from a Sharpe ratio of 0.43 to 0.94 over July 1973 to August 2026, but none of the other Fama-French factors. We explain why. Sorting stocks on past returns loads momentum on the market in proportion to the market's own return over the formation window. Using Tweedie's formula, we derive this beta in closed form from two cross-sectional quantities, the dispersion of stock betas and the density of past returns at the portfolio breakpoints, and test it on CRSP with no fitted parameter. Against realized betas from 1932 to 2025, the prediction has a slope of 0.92 (standard error 0.19) and 79% of the explanatory power of an in-sample regression, but on average it overstates the beta by 0.20, so the joint test of an exact prediction rejects. Months after market declines are 29% of the sample and supply 71% of the covariance through which timing adds to momentum's return. If the market's premium rises with its variance, the same logic implies that the market should not be volatility-timed; in our sample it is not improved by it.

Keywords: volatility-managed portfolios, momentum, time-varying beta, Tweedie's formula, factor timing

JEL: G11, G12, G14

1 Introduction

Moreira and Muir (2017) show that scaling a factor's exposure by the inverse of its recent variance raises the Sharpe ratio of many equity factors. Their evidence is a spanning regression that uses full-sample moments no investor had. In real time the gains mostly disappear (Cederburg et al., 2020; Liu et al., 2019; Barroso and Detzel, 2021), and momentum is the consistent survivor, in line with the crash evidence of Barroso and Santa-Clara (2015) and Daniel and Moskowitz (2016). Why momentum, and why not the market, for which the rule was first proposed?

We answer in three steps. First, we rebuild the six most widely used equity factors so that every position uses only past data, leverage is capped and costs are paid; only momentum improves. An

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exact decomposition shows why: timing adds return only when the factor earns less in turbulent months, and among the six factors that holds for momentum alone. Second, we derive where momentum’s turbulent-month losses come from. Momentum is long past winners and short past losers, so after a market decline its short leg holds the high-beta stocks and the factor is short the market (Grundy and Martin, 2001; Daniel and Moskowitz, 2016). Proposition 1 gives this beta in closed form from two quantities measured in the cross-section of stocks, and we test it on CRSP without fitting any parameter. Third, we show what volatility timing should time: only the part of a premium that is not paid for market risk. If the market’s premium rises with its variance (Merton, 1980), the market should not be timed at all.

The contribution is the formula and its parameter-free test. The sign of momentum’s conditional beta is known (Grundy and Martin, 2001; Daniel and Moskowitz, 2016); we derive its size and test it without estimating anything from momentum’s returns. The main test was specified in a plan committed to the code repository before the CRSP data were downloaded, and every later test before it was run (`paper/PLAN_mechanism.md`); the plan records each amendment and every negative result.

2 Data and real-time method

We use the market, size (SMB), value (HML), profitability (RMW), investment (CMA) and momentum factors of Fama and French (2015) and Carhart (1997) from the Kenneth R. French Data Library, July 1963 to August 2026. Monthly returns f_m come from the monthly files; daily returns r_d serve only to measure risk. The variance signal uses every daily return with a one-month half-life,

$$\hat{\sigma}_d^2 = \lambda \hat{\sigma}_{d-1}^2 + (1 - \lambda) r_d^2, \quad \lambda = 2^{-1/21}, \quad S_m = 21 \hat{\sigma}_{d(m)}^2, \quad (1)$$

with $d(m)$ the last trading day of month m . This learning rate was chosen for momentum on 1973 to 2016 data among 24 pre-registered variants (`paper/PLAN_adaptive.md`); a rule that switches each month to the variant with the best trailing ten-year Sharpe ratio did worse than the canonical rule (0.70 against 0.82 from July 1983, $p = 0.019$). The canonical prior-month realized variance of Moreira and Muir (2017) is reported alongside.

The exposure held in month m is set at the end of month $m - 1$:

$$w_m = \min\left(\frac{c_{m-1}}{S_{m-1}}, 1.5\right), \quad c_{m-1} = \frac{\text{sd}(f_1, \dots, f_{m-1})}{\text{sd}(f_s/S_{s-1} : s \leq m-1)}, \quad (2)$$

an expanding-window version of Moreira and Muir’s constant with their higher cap. The first 120 months are a burn-in, so positions start in July 1973 (638 months). Exposure changes cost 14 basis points per unit, their highest cost level: $f_m^\sigma = w_m f_m - \kappa |w_m - w_{m-1}|$. Sharpe ratios are compared with the studentized circular block bootstrap of Ledoit and Wolf (2008) (12-month blocks, 4,999 resamples), Holm-adjusted across the six factors (Holm, 1979).

Why timing can help or hurt. The mean managed return decomposes exactly as $\bar{f}^\sigma = \bar{w} \bar{f} + \widehat{\text{cov}}(w_m, f_m) - \kappa \overline{|\Delta w|}$. A constant exposure $\bar{w}$ has the original factor’s Sharpe ratio, so whenever

$\bar{f} > 0$,

$$\frac{\text{SR}(f^\sigma)}{\text{SR}(f)} = \underbrace{\left(1 + \frac{\widehat{\text{COV}}(w_m, f_m) - \kappa |\Delta w|}{\bar{w} \bar{f}}\right)}_{M: \text{ return channel}} \times \underbrace{\frac{\bar{w} \text{sd}(f)}{\text{sd}(f^\sigma)}}_{R: \text{ risk channel}}. \quad (3)$$

R is the risk that volatility timing removes. M is the return it adds or gives up: it exceeds one only if the factor earns more in calm months, when the rule holds more of it.

3 Only momentum improves

Factor	Managed factor, net				Real-time combination, net			Largest loss, equal volatility	
	Original	Canonical	Daily-data	p (Holm)	Original	Combination	p (Holm)	Original	Managed
Market	0.51	0.51	0.48	1.000	0.57	0.43	0.506	−68	−42
Size (SMB)	0.21	0.12	0.13	1.000	0.10	0.02	0.137	−77	−85
Value (HML)	0.32	0.23	0.27	1.000	0.24	0.18	0.343	−79	−67
Profitability (RMW)	0.38	0.47	0.45	1.000	0.47	0.35	0.545	−51	−46
Investment (CMA)	0.45	0.34	0.41	1.000	0.37	0.30	0.506	−31	−47
Momentum	0.43	0.90	0.94	0.023	0.42	0.80	0.343	−76	−23

Table 1: Annualized Sharpe ratios, July 1973 to August 2026, exposure capped at 1.5, net of 14 basis points per unit of exposure change. Canonical is the prior-month realized-variance rule; daily-data is equation (1). The combination is the real-time mix of original and managed factor of Cederburg et al. (2020), from July 1978. p -values test equal Sharpe ratios against the original factor, Holm-adjusted across the six factors; bold marks adjusted $p < 0.05$. Largest loss is the maximum peak-to-trough decline of summed monthly returns, in percentage points, at the original factor’s volatility.

Managed momentum raises the Sharpe ratio from 0.43 to 0.94 after costs ($p = 0.004$, Holm-adjusted 0.023; Table 1), and its largest loss at equal volatility falls from 76 to 23 percentage points. No other factor improves; the market’s Sharpe ratio falls from 0.51 to 0.48. No real-time combination beats its original factor after costs. Momentum’s gain is smaller after the published samples end: from 0.48 to 1.02 before 2017 and from 0.21 to 0.44 after ($p = 0.288$ over 116 months), a difference between the periods that is not significant ($p = 0.242$). Across variance signals, block lengths, caps, scaling rules and start dates, managed momentum’s full-sample Sharpe ratio ranges from 0.78 to 0.97 and always differs from the original ($p \leq 0.014$).

Table 2 shows the reason. Momentum’s covariance between exposure and return is positive in both periods, so its return channel exceeds one ($M = 1.37$ before 2017). Every other factor’s covariance is negative: the rule scales them down in the months they pay. For the market, M is 0.91 before 2017 and 0.67 after. The next two sections explain both signs.

4 Momentum’s formation beta

Proposition 1. *Let stock i ’s market beta be $b_i = \bar{b} + \sigma_b \varepsilon_i$ with $\varepsilon_i \sim \mathcal{N}(0, 1)$, and its formation-window log return $r_i = b_i R_F + u_i$, where u_i is independent of b_i and has any distribution, and R_F is the coefficient on beta, to first order $\sum_t R_{M,t}(1 - R_{M,t} - r_{f,t})$ over the window for market excess returns $R_{M,t}$. Winners are the stocks with $r_i \geq c_{hi}$ (a share q_{hi}), losers those with $r_i \leq c_{lo}$ (a share*

Factor	Period	$\bar{w}$	$\bar{f}$	$\bar{w}\bar{f}$	$\widehat{\text{cov}}(w, f)$	Cost	$\bar{f}^\sigma$	M	R
Market	1973 to 2016	0.74	6.9	5.1	-0.2	0.2	4.6	0.91	1.13
	2017 to 2026	0.73	13.1	9.6	-2.8	0.3	6.5	0.67	1.14
Size (SMB)	1973 to 2016	0.59	3.2	1.9	-0.3	0.2	1.4	0.71	1.10
	2017 to 2026	0.43	-2.6	-1.1	-0.9	0.1	-2.2	n/a	0.90
Value (HML)	1973 to 2016	0.79	4.5	3.6	-0.9	0.2	2.4	0.67	1.18
	2017 to 2026	0.27	-1.7	-0.4	-1.0	0.1	-1.5	n/a	1.08
Profitability (RMW)	1973 to 2016	0.84	3.4	2.8	-0.4	0.2	2.3	0.79	1.52
	2017 to 2026	0.40	2.4	1.0	-0.4	0.1	0.4	0.45	1.33
Investment (CMA)	1973 to 2016	0.95	4.2	4.0	-0.8	0.2	2.9	0.73	1.22
	2017 to 2026	0.50	-1.0	-0.5	-1.1	0.2	-1.8	n/a	1.11
Momentum	1973 to 2016	0.82	7.3	6.0	2.5	0.2	8.2	1.37	1.56
	2017 to 2026	0.35	2.9	1.0	1.0	0.1	1.9	1.87	1.13

Table 2: Exact decomposition of the daily-data rule, equation (3). $\bar{f}$, $\bar{w}\bar{f}$, the covariance, the cost and $\bar{f}^\sigma$ are annualized, in percent. M and R are the return and risk channels; their product is the ratio of managed to original Sharpe ratios. M is not reported when $\bar{f} \leq 0$.

q_{lo}). If betas persist into the holding month, the equal-weighted winners-minus-losers beta is

$$\beta_{WML} = \sigma_b^2 R_F \left[\frac{\phi(c_{hi})}{q_{hi}} + \frac{\phi(c_{lo})}{q_{lo}} \right], \quad (4)$$

where ϕ is the cross-sectional density of formation returns.

Proof. Write $r_i = \theta_i + e_i$ with $e_i = (b_i - \bar{b})R_F \sim \mathcal{N}(0, \sigma_b^2 R_F^2)$ independent of $\theta_i = \bar{b}R_F + u_i$. Tweedie's formula (Efron, 2011) gives $\mathbb{E}[e_i | r_i] = -\sigma_b^2 R_F^2 (\log \phi)'(r_i)$, so $\mathbb{E}[b_i | r_i] = \bar{b} - \sigma_b^2 R_F (\log \phi)'(r_i)$. Averaging over winners, $\mathbb{E}[-(\log \phi)'(r) | r \geq c_{hi}] = q_{hi}^{-1} \int_{c_{hi}}^{\infty} -\phi'(r) dr = \phi(c_{hi})/q_{hi}$; over losers the same integral gives $-\phi(c_{lo})/q_{lo}$. Subtract. $\square$

The sign comes from the market's own formation return: after a bull market momentum is long the market, after a bear market short it. With normal u_i and equal tail shares, (4) is linear in R_F near zero, the relation Grundy and Martin (2001) estimate by regression, and saturates after extreme markets. Its size depends on how many stocks sit near the breakpoints, so heavier-tailed idiosyncratic returns, which put more stocks near the 30th and 70th percentiles at the same variance, raise it.

Test. We measure both inputs in CRSP (common stocks on NYSE, AMEX and NASDAQ, monthly, to December 2025). Our value-weighted rebuild of French's momentum correlates 0.9989 with his over 1,181 months. For the portfolio formed at the end of month m , each stock's beta $\hat{b}_i$ is estimated over months $m - 71$ to $m - 12$, before the formation window starts, so no beta uses the returns being sorted on. σ_b^2 is the cross-sectional variance of $\hat{b}_i$ minus the mean of their squared standard errors, and ϕ at the NYSE breakpoints comes from the quantile-spacing estimator with the bandwidth of Hall and Sheather (1988). Both are computed within each of French's two size groups and averaged. The test regression is

$$\text{MOM}_{m+1} = a + c \hat{\beta}_m R_{M,m+1} + d R_{M,m+1} + \varepsilon_{m+1}, \quad (5)$$

with Newey-West standard errors (12 lags). The prediction is exact if $c = 1$ and $d = 0$. On simulated markets in which (4) holds, with normal and t_4 idiosyncratic returns, the procedure returns c of 1.04 on average and the joint test rejects at its nominal rate.

Sample	Portfolio	c (s.e.)	d (s.e.)	$p: c = 1, d = 0$	ΔR^2	Share of in-sample	Months
1932 to 2025	Equal-weighted	0.88 (0.20)	-0.22 (0.06)	0.000	0.21	0.75	1122
1932 to 2025	French's momentum	0.92 (0.19)	-0.20 (0.06)	0.000	0.24	0.79	1122
1963 to 2025	Equal-weighted	1.00 (0.17)	-0.16 (0.06)	0.014	0.14	0.79	750
1963 to 2025	French's momentum	1.08 (0.17)	-0.15 (0.06)	0.026	0.16	0.81	750

Table 3: Parameter-free test of Proposition 1. ΔR^2 is the increase in R^2 over a regression on $R_{M,m+1}$ alone; the share compares it with the increase from an in-sample regression that fits the slope on $R_F R_{M,m+1}$ (Grundy and Martin, 2001). Equal-weighted is the theory's portfolio, French's momentum the traded factor.

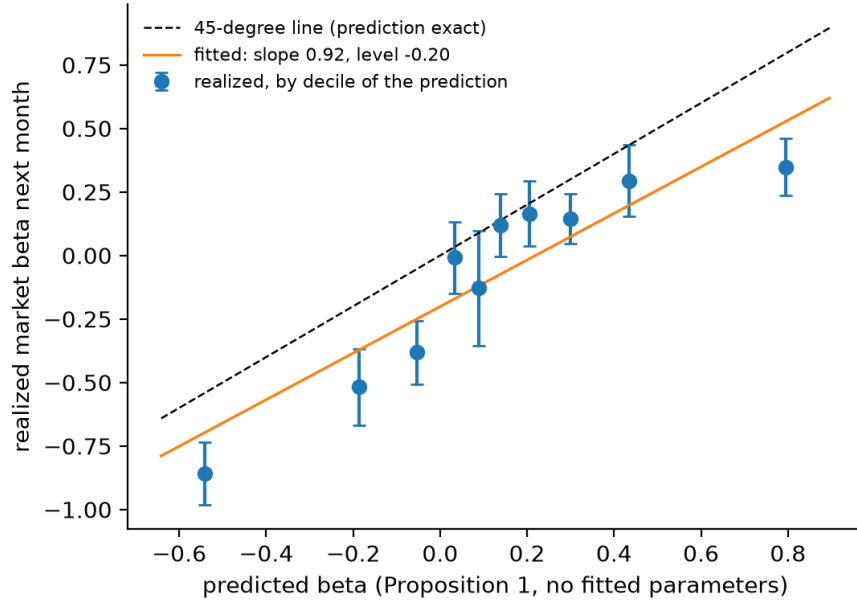


Figure 1: Realized market beta of French's momentum in the month after formation, estimated within each decile of the predicted beta (4), holding months from July 1932 to December 2025, with 95% intervals. The dashed line is the exact prediction; the solid line is the fitted relation from Table 3.

The slope matches the prediction (Table 3 and Figure 1). With no fitted parameter, c is 0.92 (standard error 0.19) for French's momentum over 1932 to 2025 and 1.08 (0.17) from 1963; for the theory's equal-weighted portfolio from 1963 it is 1.00. Adding the prediction raises the R^2 of a regression of momentum on the market by 24 percentage points, 79% of the rise from an in-sample regression that fits the slope. The level is not: d is -0.20 over the full sample and -0.15 from 1963, so the joint test rejects (from 1963, $p = 0.026$), and in the top decile the realized beta falls short of the prediction.

The level gap has a candidate explanation. Compounding lowers the formation return of volatile stocks, so if high-beta stocks are also more volatile, losers hold high-beta stocks even when $R_F = 0$. Simulated markets built that way show a negative d , but a first-order correction for it failed validation, so we report the gap as open. Measured directly from pre-window betas, the formation

beta has a slope of only 0.55 on the prediction (correlation 0.71), consistent with betas drifting between the estimation window and the formation year.

5 What volatility timing should time

Proposition 2. *If the market premium is proportional to market variance, $\lambda_t = \kappa F_t^M$ (Merton, 1980), a factor with conditional expected return $\alpha + \beta_t \lambda_t$ and variance F_t has mean-variance weight*

$$w_t^* = \frac{1}{A} \left[\frac{\alpha}{F_t} + \kappa \beta_t \frac{F_t^M}{F_t} \right]. \quad (6)$$

Inverse-variance timing is optimal only for α , the part of the premium not paid for market risk. For the market ($\alpha = 0$, $\beta = 1$, $F = F^M$) the optimal weight is constant: an inverse-variance rule cuts exposure in exactly the months that the premium rises. The market’s return channel below one in Table 2 is consistent with that. For momentum, Proposition 1 makes β_t negative after market declines, so the market-related part of its expected return is most negative when market variance is high, in the months an inverse-variance rule already avoids.

The data agree. Over 629 holding months from July 1973 to November 2025, the months after market declines ($\hat{\beta} < 0$) are 29% of the sample and contribute 71% of the covariance between momentum’s managed exposure and its return. In those months momentum earns 1.6% a year against 8.6% otherwise, and the rule holds 0.56 times the factor against 0.82. Volatility management does not need to forecast this: a mean forecast built from $\hat{\beta}_m$ alone does not beat the historical mean (out-of-sample R^2 of -0.30% before 2017, Clark-West $p = 0.154$; Clark and West, 2007). A likely reason is that it omits the level gap, whose effect on the mean scales with market variance.

6 Conclusion

Volatility management pays for momentum and for none of the other five factors, in real time and after costs. The reason is how momentum is built. Sorting on past returns makes its market beta a function of the market’s own recent return, and Proposition 1 gives that function in closed form. With no fitted parameter it predicts how momentum’s beta moves, although not its average level. After market declines the beta turns negative, and those months carry most of the covariance through which timing adds to momentum’s return. If the market’s premium rises with its variance, timing the market by variance gives up the premium; the market should not be volatility-timed, and in our sample it gains nothing.

Two limits apply. The level gap in momentum’s beta remains unexplained. We charge costs only on changes in exposure, not on the trading inside each factor portfolio, so every net figure is an upper bound.

Data availability. Factor returns are available from the Kenneth R. French Data Library; stock returns are from CRSP through WRDS and cannot be redistributed. Code and the pre-registration plans are available from the author on request.

Declaration of generative AI and AI-assisted technologies in the writing process. During the preparation of this work the author used generative AI tools to create the simulations and code

and to draft the content structure of the paper. The author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

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